

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: **Spark**

Project / Product Name: **Ripple (reference application: "Ripple for Care", a live support-agent assistant) · Samsung PRISM GenAI Hackathon 2026-27, Theme 04 Streaming Live RAG**

Organization / Institution (if any): **VIT Vellore**

Submission Date: 30 September 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? **Yes**

AI coding assistants were used throughout development: Claude Code (Anthropic, model Claude Opus 5) from 15 to 26 Sep 2026, and Claude in Cowork (Anthropic) from 26 Sep 2026 for repository setup and this form. Every commit made with AI assistance is marked with a Co-Authored-By line in the public git history. The team directed the work, reviewed it, and ran every measurement on its own machine; the failures found that way drove most of the later fixes listed below.

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming **Yes: analysed the official PRISM deck and Theme 4 Guide; generated and scored 12 product concepts against the published rubric. The team chose the Care-Call framing.**

Code generation or assistance **Yes: the engine, controller, claim graph, retrieval, evaluation harness, baselines and ablations were written with Claude Code.**

UI / UX design **Yes: the zero-build HTML dashboard (transcript, controller timeline, answer with version diff).**

Content creation **Yes: the synthetic Care Knowledge Pack corpus (60 docs / 151 chunks), the benchmark scenarios, the documentation and the submission deck.**

Data analysis **Yes: threshold calibration sweeps, ablations, and interpretation of benchmark results.**

Testing / debugging **Yes: property tests and diagnosis of bugs found by measurement (listed per feature below).**

Other **Inside the product**: the synthesis step can optionally call Google Gemini (gemini-3.8-flash) to word answers. This is off by default; the default path uses no model and no API key. The model never writes citations; they are derived from evidence IDs.

4. Feature Origin Classification (Please add/delete based on the number of the feature)

Feature 1

1. Feature Name: **Competition analysis and concept selection**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Analyse the official PRISM deck and Theme 4 Guide; generate and score product concepts against Samsung's rubric.

Output summary: 12 candidate concepts with a weighted scoring matrix; recommended a care-call copilot.

Modification: Team chose the concept and corrected the timeline assumption (10 days, not 14).

Feature 2

1. Feature Name: **M1: Retrieval-space stability controller**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Design a controller that decides when to retrieve from a partial utterance with no LLM call on the hot path.

Output summary: Rank-Biased Overlap drift / stability / novelty policy; threshold derived from a cost sweep.

Modification: Changed after measurement: shadow retrieval now runs below the minimum prefix length (early retrieval 0.74 to 0.86); EMA seeded with the first observation; suppressed turns now report retrievals that already fired, so false triggers are not hidden.

Feature 3

1. Feature Name: **M2: Claim graph and delta refinement**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Represent the answer so a late constraint patches only affected claims and citations cannot drift.

Output summary: Intent / Evidence / Claim / AnswerVersion graph; citations derived from evidence IDs, never written by a model.

Modification: Speculative drafts routed through the abstention gate after a draft was found citing a source for an unanswerable question; the delta metric was replaced with claim_survival because the original was unreadable.

Feature 4

1. Feature Name: **M3: Coverage-budgeted fusion and retrieval**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Fix sub-intent starvation that reciprocal rank fusion hides.

Output summary: Per-intent floor + marginal-gain fill + MMR over a FAISS flat index with TF-IDF+SVD embeddings.

Modification: Two AI-proposed components removed on ablation evidence: BM25 (recall 0.893 to 0.960 without it) and the reranker (coverage 0.689 to 0.724 without it, ~90 ms saved).

Feature 5

1. Feature Name: **Multi-intent decomposition and abstention gate**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Split one utterance into the questions it implies; detect questions the corpus cannot answer.

Output summary: Clause-based decomposition; abstention gate.

Modification: Two AI-proposed abstention heuristics were measured and rejected (documented in relevance.py); gate now splits compound questions first and handles hyphenated terms. Multi-intent score 0.778 to 0.889.

Feature 6

1. Feature Name: **Care Knowledge Pack corpus**
2. Self-Generated/AI-Generated/Both: **AI-Generated**
3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Author a synthetic support corpus exercising multi-intent questions, late-constraint overrides, contradictions, distractors and a deliberate coverage hole.

Output summary: 60 documents / 151 chunks with DOC_ID §Section markers. Entirely synthetic; not Samsung material.

Modification: Region scoping added after an over-invalidation bug; grown from 21 to 60 documents.

Feature 7

1. Feature Name: **Benchmark scenarios and gold labels**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Write dev and held-out scenarios with labelled turns and gold citations.

Output summary: 40 dev + 34 held-out scenarios.

Modification: Build-time check that every gold citation exists; held-out split kept unopened for a single post-freeze run.

Feature 8

1. Feature Name: **Evaluation harness, baselines, ablations and M4 signed TTFT**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Implement the baselines, all six Theme 4 gate metrics, and TTFT measured relative to end of utterance.

Output summary: Harness with B0-B3 baselines, five ablations, calibration sweeps.

Modification: Ablations changed to switch exactly one variable each; a recall measurement bug fixed; runs abort instead of tabulating results when most model calls fail.

Feature 9

1. Feature Name: **Live dashboard (frontend)**
2. Self-Generated/AI-Generated/Both: **AI-Generated**
3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Three-column live view: transcript, controller timeline, answer with version diff.

Output summary: Zero-build HTML dashboard over WebSocket.

Modification: Reviewed and run by the team; no material manual changes recorded in the commit history.

Feature 10

1. Feature Name: **LLM provider integration (Gemini) and key tooling**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Add an optional real-model provider and a key checker.

Output summary: Gemini provider, scripts/check_key.py, one-command setup.

Modification: Rewritten several times after failures observed on the team's machine: empty answers from a reasoning model, a byte-order mark in .env written by PowerShell, retired model names, and per-minute vs per-day rate limits. Guessed prices removed so cost stays zero unless a real published price is set.

Feature 11

1. Feature Name: **Docker and reproducible setup**
2. Self-Generated/AI-Generated/Both: **Both**
3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): One command on a clean machine, no key, no GPU.

Output summary: Dockerfile / docker-compose with the index built at image build time; .env loaded automatically.

Modification: Warning added when a requested model silently falls back to the keyless stub.

Feature 12

1. Feature Name: **Real-model evaluation and measurement fixes**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description

AI tool / platform: Claude in Cowork (Anthropic)

Prompt used (summary): Size a real-model benchmark run to the free-tier quota, run it, and report the results honestly.

Output summary: Full dev-split run on Google gemini-3.5-flash-lite (26 Sep 2026): 158 of 158 model-written claims supported by their cited passage; new evaluation/claim_support.py.

Modification: The team ran every benchmark on its own machine. The run exposed three measurement bugs (uncounted cost of discarded drafts, time-to-first-token excluding model time, and then including rate-limit queueing), all fixed before the reported run; the old citation-validity figure was found to be 1.000 by construction and replaced as the groundedness measure; a leaked API key was revoked.

Feature 13

1. Feature Name: **Documentation and submission deck**

2. Self-Generated/AI-Generated/Both: **Both**

3. Description

AI tool / platform: Claude Code (Anthropic), model Claude Opus 5

Prompt used (summary): Write the architecture brief, evaluation report, README, demo script and a deck on Samsung's template.

Output summary: docs/ and a 12-slide deck; every figure taken from the benchmark results file.

Modification: Reviewed by the team; unmeasured quantities are stated as unmeasured rather than estimated.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

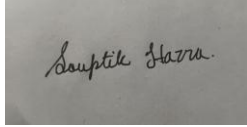
No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Souptik Hazra

Role: Team Member

Signature:

A rectangular box containing a handwritten signature in cursive script that reads "Souptik Hazra".

Date: 30 September 2026